

Question 1 — Foundations of Deep Learning (10 marks)

(a) Define deep learning. How does deep learning extend the idea of traditional machine learning?

(5 marks)

(b) Give one example where deep learning fails compared with traditional methods, and explain why.

(5 marks)

Question 2 — Optimisation & Backpropagation (10 marks)

Explain the vanishing gradient problem.

Describe (i) why it occurs, and (ii) how two different techniques help mitigate it.

Question 3 — Regularisation in Deep Learning (20 marks)

Discuss the following regularisation techniques and explain when each is most appropriate:

(a) Data augmentation

(5 marks)

(b) Early stopping

(5 marks)

(c) Label smoothing

(5 marks)

(d) Weight decay (L2 penalty)

(5 marks)

Question 4 — Neural Network Architectures (20 marks)

(a) Explain what a Residual Block is and why it improves the training of deep networks.

(10 marks)

(b) Compare Bidirectional RNNs and Transformers for sequence modelling. Include one advantage and one disadvantage for each.

(10 marks)

Question 5 — Numerical Backpropagation (20 marks)

Consider a neural network with:

Input: $x=2$

One hidden neuron with activation $f(z)=\text{ReLU}(z)$

One output neuron with activation $g(z)=\text{ReLU}(z)$

Parameters:

$w_1=0.5$ (input $\rightarrow$ hidden)

$b_1=-0.1$

$w_2=1.2$ (hidden $\rightarrow$ output)

$b_2=0.3$

Target output: $y=0.5$

Loss: $L=0.5(y - \hat{y})^2$

(a) Perform a complete forward pass and compute $\hat{y}$

(10 marks)

(b) Compute the gradients of w_1 and w_2

(10 marks)

Question 6 — Convolution & Parameters (20 marks)

A convolutional layer receives a 7×7 input with 3 channels.

It applies 4 filters of size 3×3 , stride 2, and padding 1.

(a) Compute the spatial output size.

(5 marks)

(b) How many values does the final output volume contain?

(5 marks)

(c) Compute the total number of trainable parameters (including biases).

(5 marks)

(d) Explain why increasing stride reduces computation and list one disadvantage.

(5 marks)